

Side Meeting: Taxonomy for Agentic AI Use Cases

<https://datatracker.ietf.org/doc/draft-scrm-aiproto-usecases/>

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For advice, please talk to Working Group chairs or Area Directors.

Agenda

- Welcome and Recap
- Motivation: Why agentic AI use cases matter now? Goals of the taxonomy
- Taxonomy Overview: Presentation of a proposed taxonomy; Key dimensions and design choices
- Presentations about use cases and how to apply the taxonomy
- Open discussion
- Next steps

Desired Outcome:

- Collect feedback
- Validate the taxonomy: Is it conceptually sound? Is it practical and useful?
- Identify improvements: Gaps, ambiguities, or simplifications
- Gauge interest : Who would be interested in using or contributing to it?

Taxonomy for Agentic AI Use Cases

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IETF Internet-Draft — Summary

Motivation

- Agentic AI systems — autonomous agents that communicate, coordinate, and act on behalf of users — are emerging rapidly across the industry.
- Many protocols and frameworks are being proposed, but the landscape lacks a shared structure for describing what each use case actually requires.
- This document defines a taxonomy to classify agentic AI use cases, giving the community a common vocabulary and frame of reference.
- The document does not classify use cases, it classifies capabilities required by the use cases.
- Goal: help the IETF see where existing protocols apply, where gaps exist, and which working groups might own the work.

Use Cases Requirements (1–9)

What a well-formed agentic AI use-case description should provide:

- IETF scope guidance — clarify whether and where the use case falls within the IETF's remit.
- Ecosystem boundary mapping — locate the use case relative to other SDOs and ecosystems.
- Gap analysis readiness — describe it so gaps against existing protocols can be identified.
- Adoption and layering — indicate how it builds on or layers over existing standards.
- Communication mode detail — specify the modes involved (request/response, streaming, pub/sub, etc.).
- Performance and safety needs — state the performance and safety requirements.
- Trust boundaries explicit — make the trust boundaries of the use case explicit.
- WG ownership signals — surface signals about which working group could own the work.
- Operational realism — reflect realistic operational conditions, not idealized ones.

Together these requirements make use cases comparable and ready for structured gap analysis and requirements specifications.

Use Cases Taxonomy: Seven Domains

The taxonomy organizes agentic AI use cases into seven top-level domains:

- **1. Transport** — session management, communication modes, multimodality, performance.
- **2. Security and Trust** — authentication, authorization, accountability, integrity, confidentiality, safety.
- **3. Discovery** — agent, capability, and tool discovery; service negotiation.
- **4. Identity** — naming/addressing, credentials, delegation chains, selective disclosure.
- **5. Coordination and Orchestration** — task lifecycle, agent/tool interaction, workflow orchestration.
- **6. Data and Context Management** — context exchange, provenance, knowledge, data minimization.
- **7. Operations and Management** — observability, policy enforcement, lifecycle, human-in-the-loop.

Figure 1: Agentic AI Use Cases Taxonomy

Agentic AI Use Cases Taxonomy

-
- +-- 1. Transport
 - +-- Session Management
 - +-- Communication Modes
 - +-- Content Multimodality
 - +-- Performance
 - +-- 2. Security and Trust
 - +-- Authentication
 - +-- Authorization
 - +-- Accountability
 - +-- Integrity
 - +-- Confidentiality
 - +-- Safety
 - +-- 3. Discovery
 - +-- Agent Discovery
 - +-- Capability Advertisement
 - +-- Tool Discovery
 - +-- Service Negotiation
 - +-- 4. Identity
 - +-- Agent Naming and Addressing
 - +-- Credential Management
 - +-- Delegation Chains
 - +-- Selective Disclosure
 - +-- 5. Coordination and Orchestration
 - +-- Task Lifecycle Management
 - +-- Agent-to-Agent Interaction
 - +-- Agent-to-Tool Interaction
 - +-- Multi-Agent Workflow Orchestration
 - +-- Cross-Domain Coordination
 - +-- 6. Data and Context Management
 - +-- Context Exchange
 - +-- Provenance and Citations
 - +-- Knowledge Representation
 - +-- Data Minimization
 - +-- 7. Operations and Management
 - +-- Observability
 - +-- Policy Enforcement
 - +-- Lifecycle Management
 - +-- Human-in-the-Loop

Applying the Taxonomy: A Worked Example

- The draft applies the taxonomy to an existing proposal, draft-kay-dawn-use-cases (DAWN).
- The result is a taxonomy profile: the domains and subcategories the use case touches, each annotated by role.
- Annotations distinguish a primary focus, a gap (no adequate existing protocol), and a covered (reuses existing standards).
- DAWN's primary focus is Agent Discovery; several related subcategories might be a "gap"; Accountability might be a "covered", reusing existing audit-log conventions.
- **The example shows the taxonomy's value: structured gap analysis and a clear view of where new versus existing work applies.**

Figure 2: Taxonomy Profile for draft-kay-dawn-use-cases

Taxonomy Profile: draft-kay-dawn-use-cases

```
|
+-- Transport
|   +-- Session Management ..... covered
|   +-- Communication Modes ..... gap
+-- Discovery
|   +-- Agent Discovery ..... gap (primary focus)
|   +-- Capability Advertisement ... gap
|   +-- Service Negotiation ..... gap
+-- Identity
|   +-- Agent Naming and Addressing gap
|   +-- Selective Disclosure ..... gap
+-- Security and Trust
|   +-- Authentication ..... gap
|   +-- Authorization ..... gap
|   +-- Accountability ..... covered (existing audit log conventions)
+-- Data and Context Management
    +-- Provenance and Citations ... gap
    +-- Data Minimization ..... gap
```

Historical Examples of Successful Taxonomies in the IETF

Example	Domain	Why it mattered
RFC 8199: YANG Module Classification	Network management / YANG	Created common terminology to classify YANG modules across IETF, other SDOs, open source, vendors, and users.
RFC 3040: Web Replication and Caching Taxonomy	Web caching / replication / proxies	Standardized terminology for caching, replication, proxy relationships, topologies, and replica selection.
RFC 4118: CAPWAP Architecture Taxonomy	Wireless LAN / access point control	Classified WLAN architectures based on how functions are distributed across architectural entities, helping guide CAPWAP protocol work.
RFC 7656: RTP Source Taxonomy	Real-time media / RTP	Clarified complex relationships among RTP sources and defined common terminology for protocol entities.
RFC 8406: Network Coding Taxonomy	Network coding / FEC / multicast	Provided shared terminology to avoid ambiguity in future IETF/IRTF documents on network coding.
RFC 9316: Intent Classification	Intent-based networking	Defined intent types, stakeholder perspectives, encoding approaches, and taxonomy for research and solution development.
RFC 7426: SDN Layers and Architecture Terminology	Software-defined networking	Organized SDN concepts into layers and interfaces to reduce confusion about SDN architecture.
RFC 9055: DetNet Security Considerations	Deterministic networking security	Used a taxonomy of threats and attacks to map security issues against use cases and service properties.

Example of Taxonomy: RFC 8199 YANG Module Classification

Why RFC 8199 Is a Good Analogy

- Aimed to solve a fragmented ecosystem that had no shared way to classify modules
- Set common terminology without over-constraining solutions or defining a new protocol
- Enabled consistent comparison and gap analysis across efforts
- Coordinated a broad ecosystem around a core data-modeling language

What a Taxonomy Delivers for Agentic AI

- Establishes a shared vocabulary across drafts and working groups
- Classifies use cases without forcing one architecture
- Reveals gaps, overlaps, and layering points
- Flags where protocol work may be needed
- Makes discussions concrete, repeatable, and comparable

Bottom line: A taxonomy isn't just documentation — it's a coordination mechanism that turns a broad emerging area into actionable IETF work.

Discussions, Open Questions

- Validate the taxonomy:
 - Is it conceptually sound?
 - Is it practical and useful?
- Gauge interest :
 - Interest in using it?
 - Interest in contributing to it?
- Identify improvements:
 - What are we missing?
 - Gaps, ambiguities, or simplifications

Next steps

- Collect feedback
- Update the Taxonomy Draft based on feedback
- Identify which BOFs/WGs can benefit from the taxonomy
- Continue discussions in the mailing list

Follow up questions

- **Need for concrete Agentic AI use cases**

- The current document is titled as a taxonomy for Agentic AI use cases, but it does not yet clearly describe concrete use cases.
- Several taxonomy categories, such as transport, authentication, authorization, discovery, identity, observability, and lifecycle management, appear broadly applicable to distributed systems, not uniquely Agentic AI.

- **Clarify how the taxonomy was derived**

- Reviewers asked whether the taxonomy classes were derived from a literature survey, experimentation, existing drafts, expert judgment, or another methodology.
- The document should explain the rationale behind the chosen categories and structure.

- **Consider alternative or additional classification dimensions**

- One suggestion was to classify use cases by **industrial sector**, such as manufacturing, mobility, tourism, healthcare, finance, etc.
- This could complement the current technical classification.

- **Make the taxonomy evolvable**

- The taxonomy may need to be framed as a continuously evolving document rather than a fixed classification.
- This would allow it to adapt as new use cases, protocols, and terminology emerge.

- **Coordinate with external taxonomy efforts**

- AAIF is also developing agentic AI terminology and taxonomy work, including in the Identity and Trust Working Group.
- Coordination with AAIF was highlighted as important to avoid conflicting terminology and to align use-case classification where appropriate.

- **Explain why a taxonomy is needed**

- Since many IETF agentic AI use-case drafts already exist and are written in a free-form style, the document should clearly explain the added value of defining a taxonomy.
- The key question to answer is: what problem does the taxonomy solve that the existing use-case drafts do not?

Other SDOs Efforts: AAIF Taxonomy

